

SADIQUL ISLAM SHAKIB

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SUMMARY

Final-year Computer Science and Engineering undergraduate with hands-on experience building production-grade full-stack web applications. Proficient in JavaScript, React.js, Next.js, Node.js, Express.js, and MongoDB. Experienced in architecting real-time systems using Socket.IO and WebRTC. Demonstrated ability to design scalable REST APIs, implement efficient database schemas, and deliver maintainable, well-structured codebases. Quick learner with a strong problem-solving mindset and a track record of shipping real-world projects independently.

EDUCATION

North East University Bangladesh

Bachelor of Science in Computer Science and Engineering

Universal College Sylhet

Higher Secondary School Certificate (HSC)

Sylhet, Bangladesh

July 2022 – July 2026

Sylhet, Bangladesh

Aug 2019 – Aug 2021

TECHNICAL SKILLS

Languages: JavaScript (ES6+), SQL, HTML5, CSS3

Frontend: React.js, Next.js, Tailwind CSS, ShadcnUI

Backend: Node.js, Express.js, Axios, Socket.IO, WebRTC

Databases: MongoDB, Mongoose, MySQL

Tools & Platforms: Git, GitHub, Postman, Thunder Client, Cloudinary, Vercel

Concepts: REST API Design, Real-Time Systems, Role-Based Access Control (RBAC), Pagination (Offset & Cursor), MongoDB Transactions, Server-Side Rendering (SSR), Static Site Generation (SSG)

PROJECTS

BD TravelSpirit | *Next.js, Node.js, Express.js, MongoDB, Socket.IO, Cloudinary, ML* | [Visit Here](#)

July 2025 – Present

- Architected and developed a scalable full-stack travel support platform supporting 5 distinct user roles (traveler, guide, agency, admin, moderator) with granular role-based access control (RBAC).
- Designed normalized MongoDB schemas for complex relational entities (tours, bookings, users, reviews) optimizing read/write performance and ensuring data integrity across concurrent operations.
- Implemented real-time chat functionality using Socket.IO enabling instant messaging between travelers and tour guides with message persistence and delivery tracking.
- Integrated ML-powered user analytics module to surface behavioral insights and personalize tour recommendations based on browsing and booking history.
- Engineered high-performance REST APIs supporting both offset-based and cursor-based pagination, reducing average API response time for large datasets by minimizing unnecessary document scans.
- Utilized MongoDB transactions to ensure atomicity across multi-document write operations (e.g., booking creation with simultaneous slot deduction), preventing data inconsistency under concurrent load.
- Implemented Cloudinary-based media storage pipeline for optimized image upload, transformation, and CDN delivery, reducing frontend load times for media-heavy pages.
- Built centralized error-handling middleware to standardize API error responses and simplify debugging across the entire backend service.

MeetingSync | *Node.js, Express.js, MongoDB, Socket.IO, React.js* | [Visit Here](#)

April 2025 – July 2025

- Built a real-time meeting scheduling and collaboration platform enabling users to create, update, and join meetings with live UI updates powered by Socket.IO WebSocket events.
- Designed and implemented a conflict-detection algorithm that checks for time-slot overlaps across existing bookings before confirming a new meeting, preventing double-booking scenarios.
- Optimized MongoDB queries with targeted indexing on date, time, and user reference fields to sustain consistent sub-100ms query performance under concurrent user interactions.
- Developed RESTful API endpoints for full CRUD operations on meetings, participants, and availability slots with input validation and error-handling middleware.
- Implemented session and notification management so users receive real-time alerts when meeting details are updated or new invites are received.

NextLearn | *Next.js, React.js, Tailwind CSS, REST APIs* | [Visit Here](#)

Aug 2025 – Nov 2025

- Built a production-ready Next.js learning platform demonstrating advanced rendering strategies including Server-Side Rendering (SSR), Static Site Generation (SSG), and Incremental Static Regeneration (ISR).
- Implemented dynamic routing with nested layouts and route groups, structuring the application following Next.js App Router conventions for maintainable and scalable architecture.
- Developed reusable, composable UI component library using Tailwind CSS, reducing code duplication and accelerating feature development across pages.
- Integrated Next.js API routes as lightweight backend endpoints for data fetching and mutation, eliminating the need for a separate server for internal API calls.
- Applied SEO best practices including metadata API, Open Graph tags, and semantic HTML structure, improving search engine discoverability and page ranking signals.
- Optimized application performance using Next.js Image component, lazy loading, and code-splitting, achieving improved Core Web Vitals scores.

WORK EXPERIENCE

Full-Stack Developer <i>Independent Project Development</i>	Self-Directed 2023 – Present
• Independently designed, developed, and deployed multiple production-grade full-stack web applications from concept to completion, handling frontend, backend, database, and deployment responsibilities end-to-end. • Applied software engineering best practices including modular folder structure, separation of concerns, and component-driven development to build scalable and maintainable codebases. • Built and integrated real-time communication features (live chat, video conferencing) using Socket.IO and WebRTC, gaining deep practical knowledge of WebSocket protocols and peer-to-peer connection management. • Continuously refactored and improved existing codebases to enhance code quality, reduce technical debt, and align with industry-standard patterns for REST API design and database modeling.	

LANGUAGES

Bangla: Fluent (Native) **English:** Intermediate (Professional Working Proficiency) **Hindi:** Intermediate